

JULIEN RABAULT

Applied AI / ML Engineer

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Applied AI / ML engineer. Four years at CNRS training and industrializing deep learning models in production (PyTorch, multi-GPU, Jean Zay supercomputer), with two peer-reviewed publications. Now at Berger-Levrault: building Athena, a multi-agent agentic platform (LangGraph, RAG, MCP) deployed to clients.

TECHNICAL SKILLS

Agentic & RAG	RAG / GraphRAG, tool-first agents, multi-agents, sub-agents, MCP, LangChain / LangGraph, prompt engineering, structured outputs, embeddings, fine-tuning
Deep Learning	PyTorch, Transformers, computer vision, diffusion models (DDPM / DDIM), NLP, CNN / U-Net / YOLOv8, generative models (GAN, VAE)
MLOps & Infra	Docker, AWS, CI/CD, MLflow, Airflow, Kubernetes, Celery, Langfuse, HPC / Slurm, Linux
Development	Python, FastAPI, Hugging Face, Weaviate, Mistral / OpenAI API, Git, SOLID / architecture, C#, SQL

PROFESSIONAL EXPERIENCE

AI Engineer

Jan. 2026 - present

Berger-Levrault · Toulouse · AI R&D team

Designing and building Athena, Berger-Levrault's agentic platform: agents connected to documents and business APIs for the group's clients (local government, industry, healthcare, maintenance). ~70 pilot users, Langfuse observability.

- Refactor of Athena's agentic architecture (multi-agent platform in production, LangGraph), as a team: from a router (one agent per task: RAG, MCP APIs, reports) to a single tool-first agent, with context management, skills and automatic skill creation, and sub-agent orchestration.
- Content extractor (OCR / PDF / DOCX): full rebuild, async batch processing (Celery + Mistral batch API), extensible factory/registry architecture. Deployed, -50% on extraction costs.
- MCP Builder: turns business-unit APIs into MCP servers. An LLM selects the useful endpoints, audits their gaps and generates custom tools (workflows or Python code via FastMCP), human-in-the-loop. Deployed.
- RAG ingestion chain (Airflow). Extended and added enrichment steps: data augmentation, chunking, embeddings, question/keyword generation for indexing, date filters.
- Agent evaluation and reliability: golden-set test suites, LLM-as-judge scoring and regression tracking via Langfuse.

Machine Learning Engineer

Dec. 2021 - Jan. 2026 · 4 yrs

CNRS · National AI Research Programme (PNRIA) · Toulouse

Network of AI engineers supporting research teams (weather, astrophysics, materials, ethology, biology). 10+ projects supported, up to two in parallel (6-12 months), for Météo France, CNES, CEA, INEE. Training and fine-tuning on Jean Zay (multi-GPU DDP, 8 GPUs, Slurm).

- GENS / MetScore, Météo France:** multi-GPU optimization and fine-tuning of a diffusion model (DDPM) in PyTorch; built MetScore (metrics library), still in production. Diffusion POC at -20% compute for equivalent quality. Co-author of the AMS 2025 paper.
- DeepFaune, CNRS / INEE:** fine-tuned YOLOv8 on 1.5M images (24 classes), class-imbalance handling. 93% accuracy, 3× faster inference. Peer-reviewed publication.
- BIGSF, CNES:** tech lead on the rebuild of a galactic-filament image-analysis library (U-Net): modular architecture, tests, documentation. Public toolbox.
- Others:** AUTOFILL (CEA, PairVAE, MAE 0.98), MORPHOGAN (StyleGAN2, Univ. Lorraine). Taught "Introduction to LLMs" (3h, ~25 PhD students / researchers).

Software Engineer (apprenticeship)

Aug. 2020 - Sept. 2021 · 1 yr

Agileo Automation · Montauban

Supervision and control framework for robotic machinery (semiconductors): C#, object-oriented architecture, HMI, CI/CD. Team of 5, Agile / Scrum.

EDUCATION

MSc Artificial Intelligence & Pattern Recognition (IARF), Université Paul Sabatier Toulouse III / IRIT 2019 - 2021

BSc Computer Science, Université Paul Sabatier Toulouse III 2016 - 2019

Languages: French (native) · English (full professional proficiency, scientific writing, technical documentation)

PUBLICATIONS

[Enriching Operational High-Resolution Ensemble Forecasts with StyleGAN-2](#). AIES, 2025. 3rd author, peer-reviewed.

[The DeepFaune initiative: automatic identification of European fauna](#). N. Rigoudy et al. Co-author, peer-reviewed.

OPEN SOURCE PROJECTS

LLMock (PyPI). LLM mock server: retries / fallbacks, 10+ providers, OpenAI-compatible. Python, FastAPI.

DDPM-weather. Probabilistic diffusion model for weather image denoising, -20% resources. PyTorch.

BananaML. End-to-end ML pipeline on AWS: computer vision + REST API. FastAPI, Docker, CI/CD.